Research Conclusions

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Historical statements and test counts below retain their original scope. Consult the master report for current status.

Research conclusions

Hypothesis

Explicit observations, entities, relations, states, actions, events, results, goals, memory and verification might improve learning and structural generalization, especially problem-budget efficiency. Architecture existence is not evidence for this hypothesis.

Implementation

The final system adapts one frozen local SmolLM2 backbone under four conditions: conventional answer-only, structured answer-only, trajectory-supervised procedural B and binary-feedback experiential C. Procedural policies choose numerical claims among100 candidates. Real results execute those proposals unchanged, and the verifier judges after prediction. The multiplication cursor/decomposition is hand specified. Working/episodic/concept memory and bounded hypothetical simulation are implemented; conceptual interfaces are reusable, while current concrete schemas remain decimal specific.

Observed results

Exact per-budget/seed measurements, mean±sample SD, intervals, paired tests and failure groups are in docs/final_results.md and experiments/final_smollm2_v1/. The old negative phase-4 study remains a separate frozen historical result. The fresh final dataset and larger representation/head produce higher absolute procedural accuracy, but no final target or threshold-crossing efficiency ratio is assumed from that fact.

Interpretation

The experiments compare different information regimes, not only different wrappers. B gets several state/action labels per problem; C gets bounded scalar feedback on its own proposals; answer learners get one final-answer target. B's domain scaffold decomposes the numerical task. Any gain versus answer-only therefore cannot be attributed uniquely to Mindscape, memory, verification or dreams.

The matched-information conventional local supervised policy uses B's exact state/action training information and predictor while bypassing its memory wrapper. If its predictions match B, this does not establish an independent wrapper advantage. Groundedness is an evidence-output metric: absent trajectories cannot score as grounded, regardless of exact answer accuracy. State ablations establish only their measured condition/budget effects. Replay-storage and dream claims depend on their controls, not their existence.

Under the tested model, representation, supervision and compute budgets, the experiments do not establish a data-efficiency advantage or a broad structural-generalization advantage for the Mindscape architecture. Partial success at uncorrected local procedure execution is narrower than the proposed general hypothesis. The study is a rigorous measured outcome, including limitations and failed thresholds, rather than a claim that "Mindscape works."

Limitations

One arithmetic domain, one generated final dataset, three seeds, hand-defined sequencing, unequal label/query amounts, a frozen indexed-feature language backbone and cached CPU inference limit general conclusions. Statistical significance against an answer-only condition does not remove these information differences. A second learned vertical and full language-model fine-tuning were deferred. Historical GPT/Gemini GSM8K references are not direct comparators.

Future work

Use fully matched teacher information and interaction budgets across stronger sequence learners; test learned cursor/action planning; separate pretraining from categorical features and head capacity; design explicit carry-context holdouts before data generation; improve experiential exploration without providing full target trajectories; evaluate reuse rather than merely storage; benchmark cold compute; and add one small navigation backend with generic payload schemas. Any new work should pre-register its protocol and preserve these final artifacts.